TENSORFORGE 2.0 · FINAL-FORGE · SCAM-PROOF BY DESIGN

Rakshaka AI

Catches scam SMS and WhatsApp messages in Sinhala, Tamil and Singlish, and notices when scammers change tactics before it quietly fails.

Detection | Drift monitoring | Safe fallback | Human review | Gated retraining

Scams reach Sri Lankans in their own language, and the tactics keep changing

June 2026

CBSL launches its "Be Scam Proof" campaign as digital fraud and illegal investment schemes rise

Daily FT, 3 June 2026

360+

online financial fraud incidents reported to Sri Lanka CERT in a single month

Ada Derana, 3 June 2026

12,650+

complaints to Sri Lanka CERT in 2025 about social media misuse and cyber incidents, including financial scams

Sunday Times, 1 Jan 2026

- Hit hardest: **rural communities, low-income households and senior citizens** (CBSL Governor, 2 June 2026)
- A model that is accurate on launch day can **fail silently** weeks later when scammers switch to new tactics, such as fake national digital ID (SL-UDI) activation

People who are new to digital money and receive scams in Sinhala, Tamil or Singlish

727,092

government pensioners: targets of fake "pension re-verification" calls and PIN requests

Department of Pensions, Feb 2025

1,251,522

Aswesuma welfare families: targets of fake "payment pending, update your bank details" links

Ada Derana, Aug 2026 figure

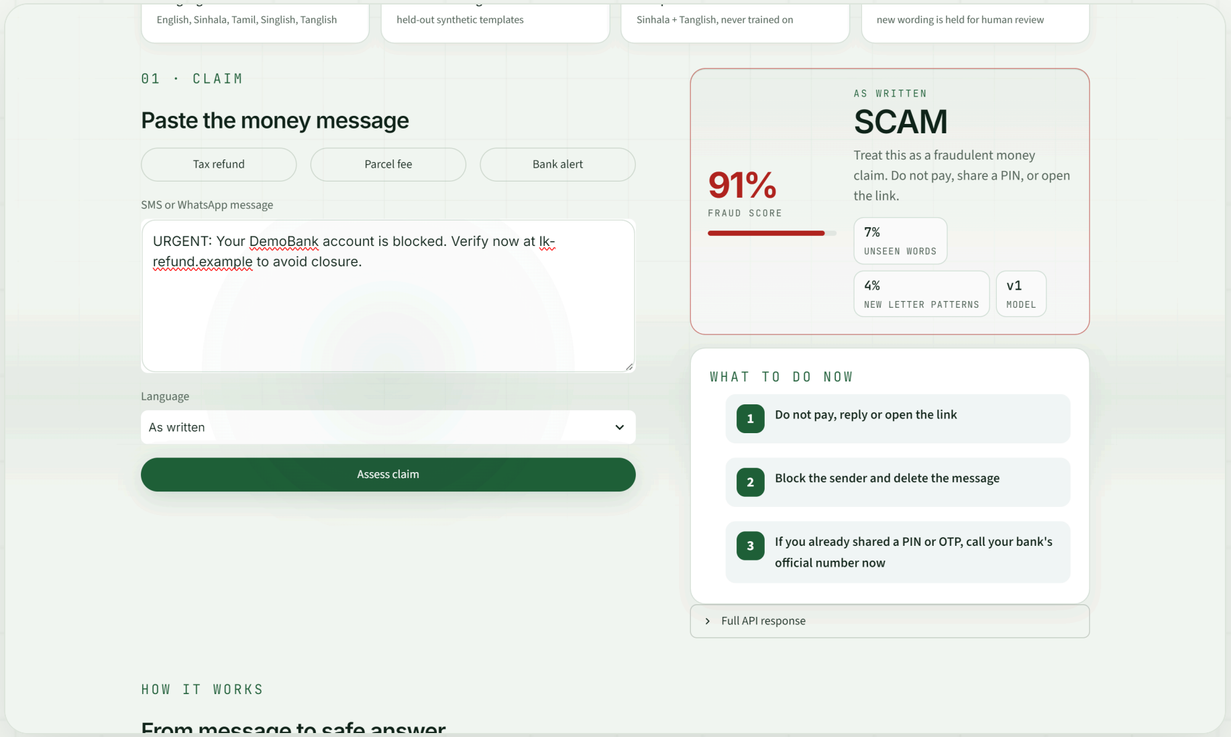
13.9M

Sri Lankans online, and 30.3 million mobile connections: every one reachable by a scam SMS

DataReportal, late 2025

First-time mobile-banking users get the same messages, with the least experience of spotting them.

A public scam checker, backed by a fraud desk that keeps it working



- **Public checker:** paste any message, get SCAM, LIKELY SAFE or UNCERTAIN, a fraud score and what to do next
- **Five languages:** English, Sinhala, Tamil, Singlish and Tanglish, with no translation step
- **Never says "safe" when unsure:** unfamiliar wording is held for a person to review
- **Admin fraud desk** (token-protected): live monitoring, drift simulation, human review, gated retraining and rollback

04 · LIVE DEMO

What you will see in the next three minutes

1

Check a scam

An urgent "account blocked" message, then a Sinhala pension scam and a family message

`SCAM · 91%`

2

Normal traffic

50 held-out messages through the drift simulation

`Model healthy`

3

Tactics change

50 new-tactic messages: SL-UDI digital ID, QR codes, crypto, "new number"

`DRIFT · 19% → 39%`

4

Review and retrain

A reviewer labels the queue. A candidate model must pass the regression gate

`v2 deployed, or rejected`

5

Back to healthy

The new tactics are now familiar, and a bad update can be rolled back

`Model healthy`

All messages in the demo are synthetic: fake banks, 07X-XXXXXXX numbers and .example links.

Drift sends messages to people, and their labels retrain the model, behind a safety gate

Message in
Public POST /predict, rate-limited per client

→

Classifier
TF-IDF on letter chunks and words, logistic regression

→

Verdict and log
SQLite: hash and metadata; raw text only if sent to review

→

Drift monitor
Last 50 checks vs validation baseline: unseen words +12 pts or confidence −0.10

↓

Safe fallback
Verdict becomes UNCERTAIN, never "safe"; alert logged once

←

Human review
Admin reviewer sets every label; unlabelled text erased after 7 days

←

Regression gate
Candidate must not lose more than 1 pt of balanced accuracy, scam or legitimate recall

←

Deploy or reject
Hot reload with no restart; previous model kept for one-click rollback

Our own synthetic data, split so the test only contains wording the model never saw

SET	MESSAGES	PURPOSE
Train	1,365	Templates + 188 hand-written messages
Validation	276	Drift baseline only
Held-out test	264	Unseen templates
Independent	65	Written separately, never trained on
Shifted	152	New tactics for the drift demo

- **Split by template:** every message from one template lands in one split, so scores are honest
- Templates follow tactics named by CBSL and police: OTP theft, parcel fees, fake Aswesuma payments, lottery wins
- **No real personal data** (PDPA No. 9 of 2022): fake banks, numbers and links
- Every source and generation method, including AI tools, is listed in the README

07 · RESULTS

No scams missed on the held-out test set

0 / 190

scams missed on the held-out test set (macro F1 **0.946**)

11 of 74 legitimate messages flagged

0.984

macro F1 on the independent set, written separately and never trained on

1 of 30 scams missed

1.00

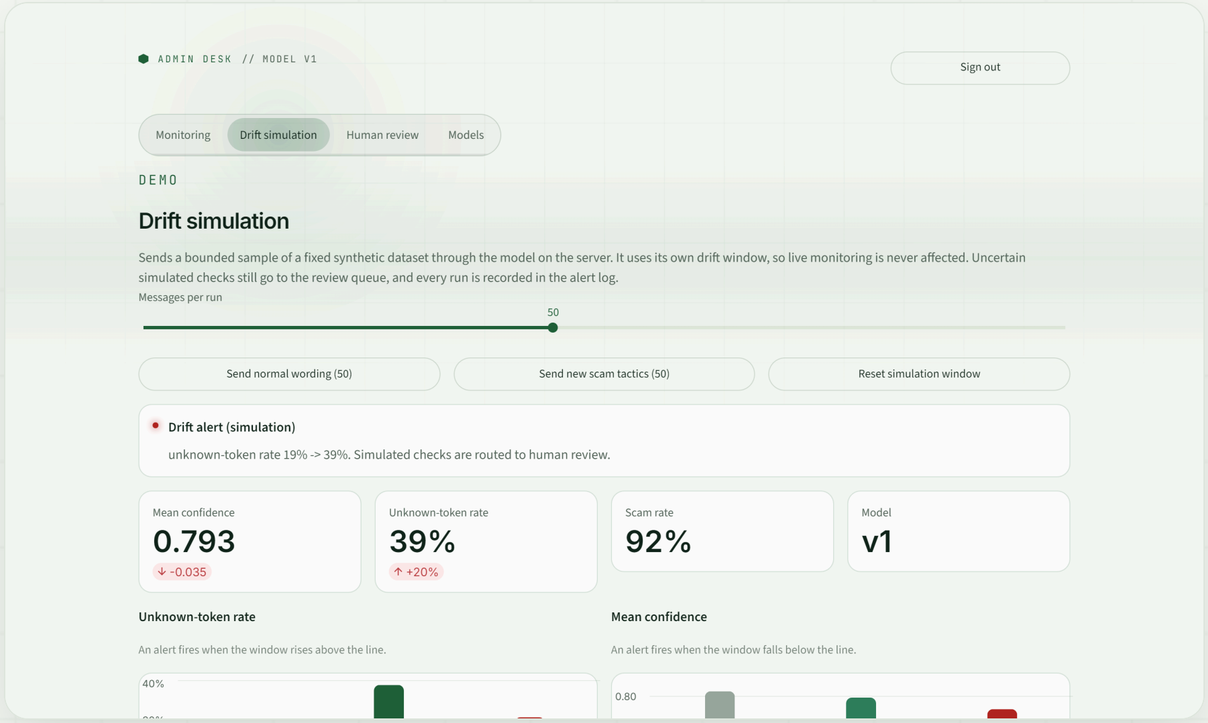
balanced accuracy on held-out English, Singlish and Tamil

scikit-learn, CPU, trains in seconds

Our known gap: Sinhala legitimate messages

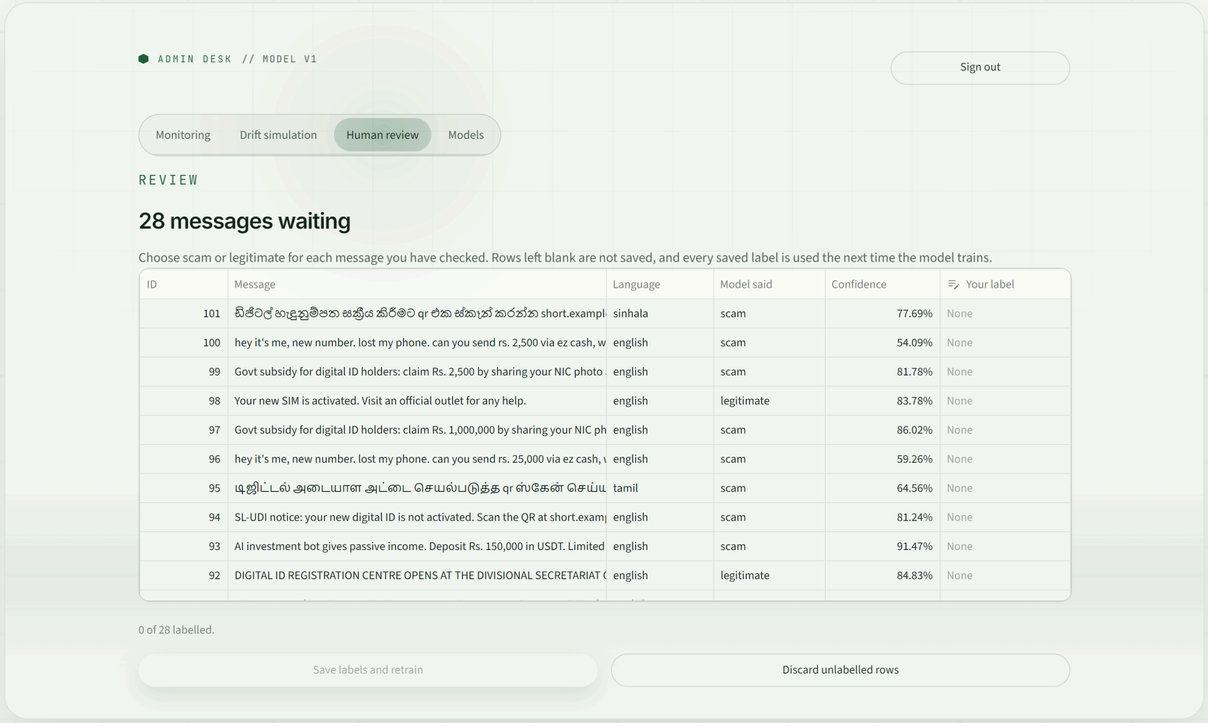
Sinhala balanced accuracy on the held-out test is **0.63**: some genuine Sinhala messages are flagged as scams. The fix is more Sinhala legitimate data, checked by native speakers. All scores come from synthetic data, not real traffic.

The alarm rings before accuracy drops



- **Signal:** share of words never seen in training, averaged over the last 50 checks, plus mean confidence
- **Calibrated, not guessed:** in 1,000 random batches, normal traffic never rose above −1 point; new tactics always rose at least +13.5. Alert line: **+12**
- **Demo:** new tactics push unseen words from **19% to 39%** while the model is still right
- Simulation runs in its own window, so live monitoring is never polluted

Fall back safely, ask a person, retrain only if it is better



- **Safe fallback:** during drift every verdict is UNCERTAIN and the message is queued
- **Human review:** reviewers set each label themselves; nothing is labelled by default
- **Regression gate:** the new model is tested against the live one on 535 fixed messages and deployed only if it is not worse
- **Rollback** in one click, and every drift, retrain and rollback is written to the alert log

Anyone can start it from the repository

Deploy

FastAPI service and Streamlit desk. **docker compose up --build**, tested by the team, or documented commands for Windows, macOS and Linux

Secure by default

Admin routes need a token and are disabled without one. Public checks are rate-limited. Message text is erased after 7 days

Reproducible

Pinned dependencies, fixed seed, 48 unit tests, CI workflow, and data, training and evaluation scripts in the repo

Monitoring is built in: a prediction log, drift endpoints, alert history and charts on the admin desk.

From one bank helpline to every pensioner help desk

MONTHS 0–6 · PILOT

A "forward a message to check it" WhatsApp number with one bank's fraud helpline. The bank's fraud team are the reviewers. **Target: 10,000 users**

MONTHS 6–12 · EXPAND

Pensioner and Aswesuma help desks at Divisional Secretariats. Aligned with CBSL "Be Scam Proof"; new tactics shared with SLCERT

SCALE

An API for banks and telcos: checks inside banking apps and at SMS gateways

How we measure harm prevented

Scams flagged before the user acted · time to detect a new tactic (one 50-message window) · false-alarm rate · new tactics reported to SLCERT

What it takes to go from demo to pilot

- **Real data:** replace synthetic training data with consented, reviewed reports from the pilot partner
- **Native-speaker review** of Sinhala, Tamil and Singlish data, starting with Sinhala legitimate messages
- **Better model:** fine-tune a multilingual transformer (for example XLM-RoBERTa) once there is enough real data
- **Operations:** page the fraud team on drift, persist the drift window, and host the service for the pilot
- **Privacy:** consent before a reviewed message is used for training, under PDPA No. 9 of 2022

SCAM-PROOF BY DESIGN

Rakshaka AI

> **Catches deception today, and notices when the scammers change tactics tomorrow.**

Repository: github.com/geemalfernando/Rakshaka_AI · AI tools used: Claude, Claude Code, ChatGPT (declared in the README) · All demo data is synthetic.